1) $\log_2(x^2-4x) = \log_2(x(x-4)) = \log_2(x) + \log_2(x-4)$

2) $\log_3\left(\frac{1}{81\sqrt{3x^2+1}}\right) = \log_3\left(81\sqrt{3x^2+1}\right)^{-1} = -\log_3\left(81\sqrt{3x^2+1}\right) = -\left(\log_3(81) + \log_3\left(\sqrt{3x^2+1}\right)\right) = -\left(\log(3^4) + \log_3(3x^2+1)^{\frac{1}{2}}\right) = -\left(4\log_3(3) + \frac{1}{2}\log_3(3x^2+1)\right) = -4-\frac{1}{2}\log_3(3x^2+1)$

3) $\log_2\left(\frac{xy}{z^2}\right) = \log_2(xy) - \log(z^2) = \log_2 x + \log_2 y - 2\cdot\log_2 z$

4) $\ln\frac{2x\sqrt{x+1}}{e^2(2x-3)^5} = \ln(2x\sqrt{x+1}) - \ln(e^2(2x-3)^5)$

8) $\log_4(x+3) + \log_4(2-x) = 1$

$\Downarrow$

$\log_4((x+3)\cdot(2-x)) = 1$

$\Downarrow$

$(x+3)\cdot(2-x) = 4^1$

$\Downarrow$

$2x - x^2 + 6 - 3x = 4^1$

$\Downarrow$

$-x^2 - x + 6 = 4$

$\quad -4 \quad -4$

$\Downarrow$

$-x^2 - x + 2 = 0$

$\Downarrow$

$(x+2)(-x+1) = 0$

$x+2=0$ ou $-x+1=0$

$x=-2$ $\quad$ $x=1$

6) $\ln(x+1) = 2$

$\Downarrow$

$e^2 = x+1$

$\Downarrow$

$e^2 - 1 = x$

4) $5^{x-3} = 3^{2x+3}$

$\Downarrow$

$\log_3(5^{x-3}) = \log_3(3^{2x+3})$

$\Downarrow$

$(x-3)\cdot\log_3(5) = 2x+3$

$\Downarrow$

$x\cdot\log_3(5) - 3\cdot\log_3(5) = 2x+3$

$\Downarrow$

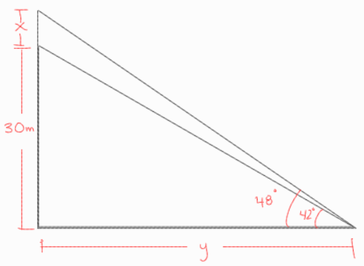
$-3\cdot\log_3(5-3) = 2x - x\cdot\log_3(5)$

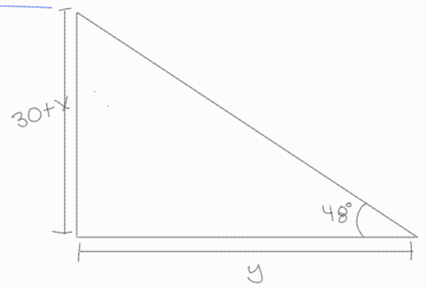
$\Downarrow$

$-3(\log_3(5+1)) = x(2-\log_3(5))$

Fonction Trigonométrique

Ex #2 Chap 8 diapo 5





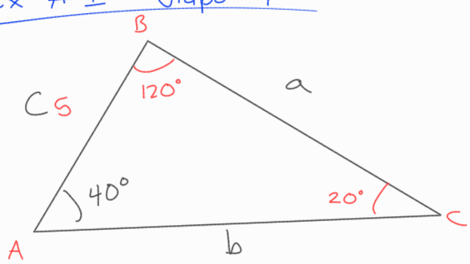
$\tan(42°) = \frac{30}{y}$

solve$\left(y = \frac{30}{\tan(42°)}\right) = 33{,}32\ m$

solve$\left(\tan(48) = \frac{30+x}{33{,}32}\right)$

$x = 7\ m$

Ex #1 diapo 7



$\boxed{\frac{a}{\sin(A)} = \frac{b}{\sin(120)}} = \frac{5}{\sin(20)}$

$b = 12{,}66$

$a = 9{,}40$

Exercice #3 diapo 10

$$\frac{11\pi}{8} \longrightarrow \text{degrés} = \frac{1980}{8} = 247{,}5°$$

$$\frac{-17\pi}{9} = \frac{-3060}{9} = -340°$$

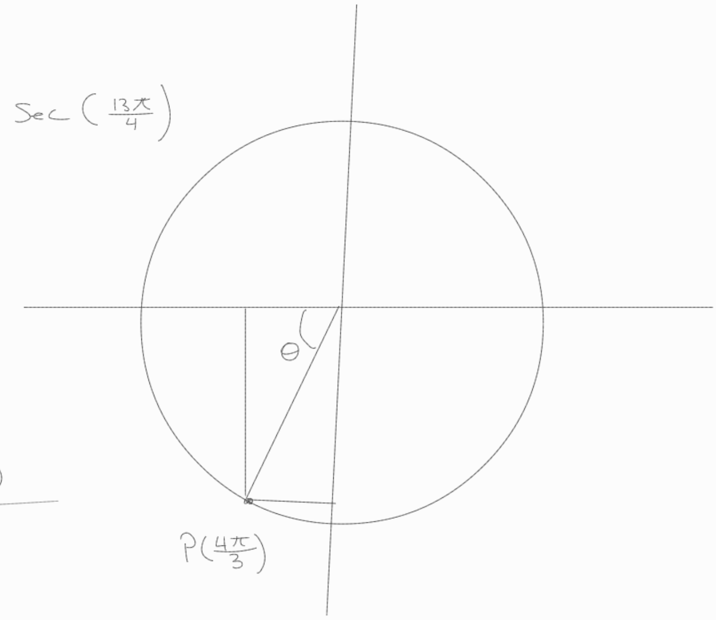
$480° \rightarrow$ rad $\qquad \frac{480}{180} = \frac{8}{3}\cdot\pi \cdot \frac{8\pi}{3}$

$-140° \rightarrow$ rad $\qquad \frac{-140}{180} = \frac{-7}{9}\cdot\pi = \frac{-7\pi}{9}$

c) $r = 15\text{cm}$ $\quad$ arc $= 45\text{cm}$

$45 = 15\cdot\theta \qquad \theta = 3\text{rad}$

$$\sec(585°) = \sec\left(\frac{585}{180}\cdot\pi\right) = \sec\left(\frac{13\pi}{4}\right)$$



$$\cot(4\pi/3) = \frac{1}{\tan(4\pi/3)}$$

$$= \frac{1}{\frac{\sin(4\pi/3)}{\cos(4\pi/3)}}$$

$$= \frac{\cos(4\pi/3)}{\sin(4\pi/3)}$$

$$= \frac{\frac{-1}{2}}{-\frac{\sqrt{3}}{2}}$$

$$= \frac{-1}{2}\cdot\frac{-2}{\sqrt{3}} = \frac{\cancel{2}}{\cancel{2}\cdot\sqrt{3}} = \frac{1}{\sqrt{3}}$$

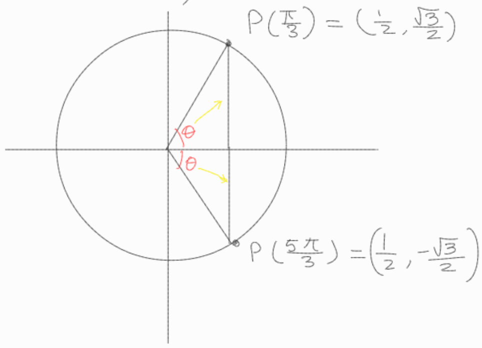
Exercice 4 ou plus

$$\arcsin\left(\frac{\sqrt{3}}{2}\right) = \sin^{-1}\left(\frac{\sqrt{3}}{2}\right) = \theta$$

on cherche θ $\left(\frac{-\pi}{2} \le \theta \le \frac{\pi}{2}\right)$

$$\sin(\theta) = \frac{\sqrt{3}}{2}$$

$$\theta = \frac{\pi}{3}$$



autre exercice diapo 20

$\text{Arc}\cos 0 = \cos^{-1}(0) = \theta$

$\cos(\theta) = 0$

$\theta = \frac{\pi}{2}$

$\arccos\left(\frac{1}{2}\right) = \theta \Rightarrow \cos^{-1}\left(\frac{1}{2}\right) = \theta$

$\Rightarrow \cos(\theta) = \frac{1}{2}$

$\theta = \frac{\pi}{3}$

$\cos\left(\arccos\left(\frac{\sqrt{3}}{2}\right)\right) = \theta \Rightarrow \cancel{\cos(\cos^{-1}}\left(\frac{\sqrt{3}}{2}\right)) = \theta$

$\frac{\sqrt{3}}{2} = \theta \qquad (f \circ f^{-1})(x) = x$

Exercice P.22 diapo

$\arctan(1) = \theta \Rightarrow \tan^{-1}(1) = \theta$

$\tan(\theta) = 1$

$\theta = \frac{\pi}{4}$

$\cos(\arctan(1)) = \cos\left(\frac{\pi}{4}\right) = \frac{\sqrt{2}}{2}$

$\arctan\left(\frac{1}{\sqrt{3}}\right) \Rightarrow \tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = \frac{\pi}{6}$